Prognosis of choroidal melanoma and the result of ruthenium brachytherapy combined with transpupillary thermotherapy in Korean patients

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ABSTRACT

Background/aims To evaluate the clinical features and prognosis of choroidal melanoma in Korean patients and the results of treatment with ruthenium (Ru) 106 plaque brachytherapy.

Methods The medical charts of 111 patients diagnosed with choroidal melanoma who underwent Ru brachytherapy with trans-pupillary thermotherapy or local resection (61 patients) or who underwent other treatments (26 primary enucleations, 22 γ-knife radiotherapies and two lamellar sclerouvectomies) were reviewed retrospectively.

Results The mean largest basal diameter (LBD) was 11.4±3.2 mm and the mean apical height was 7.8±2.9 mm. Compared with the collaborative ocular melanoma study, mean tumour height was skewed toward higher values (6.2 vs 4.8 in medium tumours, $p<0.0001$; 10.9 vs 9.5 mm in large tumours, $p=0.034$) and the LBD in large tumours was skewed toward smaller values (13.6 vs 17.3, $p<0.0001$). The estimated 5-year metastasis-free rate was 73.9% and the disease-specific survival rate was 84.6%. For the 61 patients that were treated with Ru brachytherapy, the 5-year metastasis-free and disease-specific survival rates were 79.0% and 87.7%, respectively, and the 5-year incidence of enucleation was 25.4%. The mean tumour regression at 6, 12 and 18 months after brachytherapy was 80.2%, 73.1% and 69.2%, respectively.

Conclusions Choroidal melanomas in Korean patients tend to grow vertically with a relatively large apical height and a small LBD. The prognosis of choroidal melanomas overall as well as prognosis after Ru brachytherapy were similar to those seen in previous studies with Caucasian patients. The enucleation rate after brachytherapy seems to be higher in Korean patients, for which a greater initial tumour height seems to be partly responsible.

INTRODUCTION

Choroidal melanoma is the most common primary intraocular malignancy in adults; however, its incidence differs across racial and ethnic groups. A recent study determined that the age-adjusted incidence of choroidal melanoma is 5.1 per million and that this has remained unchanged from 1973 to 2008.[1] There are few reports investigating the incidence of choroidal melanoma in Asian subjects, especially in East Asians like Koreans, Japanese and Chinese. Kaneko reported the incidence of choroidal melanoma in Japan to be 0.25 cases per million per year: only 5% of that observed in the USA.[2] Hu and colleagues estimated that the incidence of choroidal melanoma in Shanghai seems to be 37-fold lower than in New York State, suggesting that the incidence may be very low in East Asian populations.[3] Despite the low incidence rates of choroidal melanoma in Asian populations, the malignancy usually affects younger patients, and tumours tend to be larger and exhibit epithelioid cellularity more often, which result in a poorer prognosis in Asian patients than in Caucasians.[4 5] However, to date, there have been neither any large studies investigating the prognosis of choroidal melanoma nor any reports describing the outcome of brachytherapy in Asian populations.

In this study, we investigate the clinical features and prognosis of choroidal melanoma, and the results of ruthenium (Ru) 106 plaque brachytherapy in Korean patients.

MATERIAL AND METHODS

Patients and data collection

Patients selected for this study had been diagnosed with medium to large choroidal melanomas without direct optic nerve invasion between September 1993 and October 2010, and had follow-up data for a minimum of 1 year at the Ophthalmology Department of Yonsei University College of Medicine (Korea). This investigation was approved by the Institutional Review Board. Baseline clinical variables, including age, gender and laterality, were recorded. Tumour data were also collected, including location, size and distance from the posterior border of the tumour to the foveola and optic disc. The largest basal diameter (LBD) and apical height of the tumours were measured by B-scan ultrasonography, and the tumour size was classified according to collaborative ocular melanoma study (COMS) criteria.

Treatment

A total of 111 patients were included in this study. Among them, 61 patients were treated with Ru brachytherapy combined with trans-pupillary thermotherapy (TTT) or local resection, 26 patients with primary enucleation, 22 patients with γ-knife radiotherapy (GKR) and two patients with lamellar sclerouvectomy.

Before the introduction of Ru-106 plaques (Eckert & Ziegler BEBIG, Berlin, Germany) to our institute in October 2006, GKR using a Leksell γ-knife (Elekta Instruments AB, Stockholm, Sweden) was the only eye-sparing therapy for choroidal melanoma available. After Ru-106 plaques became available, we primarily performed brachytherapy as an eye-sparing therapy for choroidal melanoma. The target dose to the tumour apex was 85 Gy, while the dose was reduced in some cases in order to not exceed a sclera dose of 1000 Gy.

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All patients treated with brachytherapy without excision received adjunctive TTT. The first TTT procedure was performed at the time of plaque removal with an infrared diode laser using a slit-lamp delivery system. The exposure time was 1 min per spot and laser power was adjusted stepwise until the tumour surface turned slightly grey. On follow-up, one (5 patients) or two (35 patients) additional TTT procedures were performed in an office setting at intervals of 3 months.

Enucleation was offered as he principal treatment if the patient had a large tumour. However, for patients who strongly refused to have enucleation, local resection of the tumour was performed. If the tumour was mainly located in the ciliary body, a transscleral tumour resection was performed; if the tumour had a posterior location with a basal diameter of less than 15 mm, endoresection by pars plana vitrectomy was performed. Adjuvant brachytherapy was applied in almost all patients who received local resection treatment.

Follow-up and assessment of death

Follow-up visits were planned 3, 6, 9 and 12 months after treatment and then semiannually after this. At each visit, best-corrected visual acuity tests, slit-lamp examinations, indirect ophthalmoscopy, ultrasonography and fundus photography were administered. All patients were also given liver function tests, a chest x-ray and an abdominal sonography annually to screen for metastasis. When positron emission tomography (PET) technology became available in 2003, biannual screenings for metastasis were performed, with whole-body PET and the liver function tests/chest x-ray/abdominal sonography alternating. When metastasis was suspected from imaging or a physical examination, we consulted an internist for confirmation.

To verify the vital status of each patient up to December 2011, we contacted the National Health Insurance Corporation. Dates and causes of death were confirmed by the National Health Statistical Office and National Registry Program.

Statistical methods

The data were analysed using SPSS 17.0 (SPSS Inc, Chicago, Illinois, USA). Kaplan–Meier survival curves were used to analyse the rate of metastasis, survival and risk of enucleation. Overall survival was defined as the time elapsed from the date of the first treatment to the date of death or, in surviving patients, to December 2011. The relationship between these outcomes and the primary treatment modality were evaluated using a log-rank test.

Univariate and multivariate Cox proportional hazard regressions were performed to evaluate which factors (age, sex, primary treatment modality, tumour height, LBD, ciliary body involvement, distance from the posterior margin of the tumour to the fovea and optic disc, and local recurrence) affected metastasis, survival or secondary enucleation after treatment. The variables that were significant on a univariate level ($p<0.05$) were entered into a multivariate analysis using a stepwise method. For variables that showed a high degree of correlation, only one variable was entered into multivariate models (out of tumour LBD, height and size classification, height was entered). Tumour height and LBD were compared with the COMS study by χ^2 test or Fisher's exact test.

RESULTS

Patients and treatment characteristics

Among 111 patients included in this study, 58 (52.3%) were male. All the patients were Korean and had a mean age of 51.9±13.7 years at time of treatment (range, 24–88 years). The mean LBD was 11.4±3.2 mm (range, 3.24–19.3 mm) and the mean height was 7.8±2.9 mm (range, 2.5–15.6 mm). No patients had any clinically detectable metastasis before treatment. The patients' demographics, tumour characteristics and classification according to the treatment methods are listed in table 1.

To show the baseline characteristics of choroidal melanoma in Korean patients, we compared tumour size with COMS study data. We compared medium-sized tumours with data from the COMS medium tumour study[6] and large-sized tumours with data from the COMS large tumour study (table 2).[7] Tumour height was skewed toward higher values (mean: 6.2 vs 4.8 mm in medium tumours, $p<0.0001$; 10.9 vs 9.5 mm in large tumours, $p=0.034$). LBDs in large tumours were skewed toward smaller values (13.6 vs 17.3 mm, $p<0.0001$). However, the distribution of the LBD in medium tumours was not significantly different (10.4 vs 11.4 mm, $p=0.181$).

Survival and metastasis analysis

During the investigation period, 23 patients had distant metastasis after a mean period of 26.8±20.8 months (range, 8–81 months) following treatment. The estimated 5-year metastasis-free rate was 73.9% (95% CIs, 63.1 to 84.7). There were 15 melanoma-related deaths and two deaths from other causes (cholangiocarcinoma and gastric cancer). The median time from the first detection of distant metastasis to metastatic death was 4.8 months (range, 1–15 months) and the estimated 5-year disease-specific survival rate was 84.6% (95% CI 76.8 to 92.4).

In the univariate analysis, the factors predicting metastasis and metastatic death were LBD, height and ciliary body involvement. In the final multivariate model, the most significant predictive factor for metastasis and metastatic death was height ($p=0.012$ and 0.049) (tables 3 and 4). There was no significant difference in metastasis and survival rate according to the first treatment modality. ($p=0.142$ and 0.159, log-rank test).

Outcomes after Ru brachytherapy

Of the 61 patients treated with Ru brachytherapy, 40 patients were treated in combination with TTT, while 21 patients were treated in combination with local resection. Forty tumours treated with brachytherapy and TTT received 70.1±26.6 Gy to the apex and 719.6±239.3 Gy to the sclera. Thirteen patients had a transscleral tumour resection and eight patients had an internal choroidectomy. Of 61 patients, 11 experienced metastasis at a mean period of 21.5±8.2 months (range, 10–45 months) following the first treatment and six patients died at a median of 3 months (range, 1–17) after diagnosis with metastasis. One patient died due to gastric cancer after diagnosis with choroidal melanoma. The 5-year metastasis-free rate, disease-specific survival rate and overall survival rate were 79.0% (95% CI, 67.3 to 90.8), 87.7% (95% CI, 77.8 to 97.7) and 84.0% (95% CI, 72.2 to 95.8), respectively.

Univariate analysis showed that the LBD, height and secondary enucleation were significantly correlated with metastasis. The LBD and height were correlated with metastatic death by univariate analysis. The difference in treatment method (brachytherapy+TTT vs brachytherapy+tumour excision) showed no significant effect on the 5-year metastasis-free rate ($p=0.166$) or survival rate ($p=0.815$). There was no significant variable that was predictive for metastasis or metastatic death by multivariate analysis.

Local recurrences were diagnosed in seven patients after a median time of 21 months (range, 8–36) and the 5-year local recurrence rate was 16.9% (95% CI 4.6 to 29.2). Among these seven patients, six patients had an initial tumour height of over

Table 1 Demographic and clinical features of patients with choroidal melanoma classified by treatment modality

	Brachy	1° Enu	GKR	LSU	Total	p value
N	61	26	22	2	111	
Age, years (SD)	51.2 (13.6)	57.3 (13.7)	47.4 (12.7)	54.5 (9.2)	51.9 (13.7)	
Men/women	33/28	12/14	12/10	1/1	58/53	
Tumour size						
LBD	11.4 (2.8)	12.9 (3.4)	10.0 (3.6)	10.5 (1.2)	11.4 (3.2)	
Height	7.4 (2.8)	9.9 (2.8)	6.2 (2.0)	6.0 (2.9)	7.8 (2.9)	
COMS criteria						
Large	14	21	3	0	38	
Medium	47	5	19	2	73	
Anterior margin						
Ciliary body	12	6	3	1	22	
Ora serrata to equator	24	8	8	1	41	
Posterior to equator	25	12	11	0	48	
Posterior margin						
Ora serrata to equator	10	3	0	1	14	
Posterior to equator	51	23	22	1	97	
Distance from posterior tumour border to foveola (mm)						
0–2.9	16	4	5	0	25	
3–5.9	8	13	3	0	24	
≥6.0	37	9	14	2	62	
Distance from posterior tumour border to edge of optic disc (mm)						
<2.0	7	9	5	0	21	
≥2.0	54	17	17	2	90	
Classification (AJCC/UICC, 7th edition), n (%)						
Stage 1	11	0	8	0	19	
Stage 2	34	15	10	1	60	
Stage 3	6	4	1	0	11	
Stage 4	10	7	3	1	21	
5-year metastasis free rate (%)	79.0%	67.9%	85.6%	ND	79.3%	0.142*
5-year disease specific survival rate (%)	87.7%	77.2%	90.7%	ND	86.1%	0.126*
5-year overall survival rate (%)	84.0%	77.2%	90.7%	ND	84.6%	0.073*

*Log-rank test.
1° Enu., primary enucleation; AJCC/UICC, American Joint Cancer Committee/Union Internationale Contre le Cancer; Brachy, brachytherapy; COMS, collaborative ocular melanoma study; GKR, γ-knife radiotherapy; LBD, largest basal diameter; LSU, lamellar sclerouvectomy.

7 mm, and all seven eyes with local recurrence were enucleated. Thirteen patients eventually underwent enucleation because of neovascular glaucoma in three cases, phthisis in two cases and sympathetic ophthalmia in one case. The 5-year incidence of enucleation was 25.4% and enucleations were performed a median of 14 months (range, 6–43) after brachytherapy. Tumour recurrence and height were significant risk factors for enucleation as determined by univariate analysis. Tumour recurrence was not a significant prognostic factor for metastasis or metastatic death (table 3).

Tumour regression after Ru brachytherapy

We also investigated the ultrasonographic changes in the tumour height of 40 patients who underwent brachytherapy plus TTT without tumour excision. Before brachytherapy, the median tumour height was 5.9 mm (range, 2.5–9.9 mm). Thirty-seven (93%) patients responded to brachytherapy by exhibiting a decrease in height. Figure 1 shows the decrease in height after treatment. The mean tumour regression (ie, percentage of the original height) at 6, 12, and 18 months was 80.2%, 73.1% and 69.2%, respectively. Only two patients had complete tumour regression 18 months after treatment.

DISCUSSION

To our knowledge, this is the first study to report the prognosis of choroidal melanoma treated with brachytherapy and is the largest study to report the prognosis of choroidal melanoma within an Asian population.

The previously reported mean age at diagnosis for choroidal melanoma in Caucasians was between 57 and 60 years.[6–9] In this study, the mean age was 51.9 years and we can confirm that choroidal melanoma develops in younger patients in Asian populations.

There have been reports on the size of choroidal melanomas in Hispanic and South Asian patients, stating that LBD was larger than in Caucasians.[3 10] In our study, the mean LBD was 11.4 mm and the mean height was 7.8 mm, smaller values than in these two studies. In comparison to the COMS study, we observed that medium and large choroidal melanomas have relatively large heights with small LBDs in large melanomas. We calculated the mean height-to-base ratio to be 0.70, which is smaller than the 0.6 reported by Devron.[11] Based on this information, it appears that choroidal melanomas in Koreans have a tendency to grow vertically rather than horizontally. The relatively larger height could be indicative of the late diagnosis of choroidal melanoma in Korean patients. In this study, only 10 tumours (9.0%) were incidentally discovered during routine

Table 2 Tumour size characteristics of 111 patients diagnosed with a medium/large choroidal melanoma (according to collaborative ocular melanoma study (COMS) criteria), compared with COMS data

Large sized tumour				Medium sized tumour			
Size in mm	YUMC (N=38) n (%)	COMS data (N=1003) n (%)	p value	Size in mm	YUMC (N=73) n (%)	COMS data (N=1317) n (%)	p value
Apical height, no. (%) of patients			0.034*	Apical height no. (%) of patients			<0.0001†
≤5.0	0	90 (9)		2.5–3.0	2 (3)	162 (12)	
5.1–7.0	1 (3)	109 (11)		3.1–4.0	12 (16)	430 (33)	
7.1–9.0	5 (13)	230 (23)		4.1–5.0	9 (12)	243 (18)	
9.1–11.0	15 (39)	281 (28)		5.1–6.0	16 (22)	154 (12)	
11.1–13.0	13 (34)	191 (19)		6.1–7.0	8 (11)	139 (11)	
13.1–15.0	2 (5)	73 (7)		7.1–8.0	10 (14)	114 (9)	
≥15.1	1 (3)	28 (3)		8.1–10.0	16 (22)	75 (6)	
NA	1 (3)	1 (0)		NA	0	1	
Mean	10.8	9.5		Mean	6.2	4.8	
Largest basal diameter, no. (%) of patients			<0.0001*	Largest basal diameter, no. (%) of patients			0.1810†
≤13.0	16 (42)	78 (8)		≤8.0	13 (18)	185 (14)	
13.1–14.9	4 (10)	82 (8)		8.1–10.0	20 (28)	275 (21)	
15.0–16.9	8 (21)	215 (22)		10.1–12.0	18 (24)	371 (28)	
17.0–18.9	8 (21)	334 (34)		12.1–14.0	17 (23)	276 (21)	
19.0–20.9	1 (3)	183 (18)		14.1–16.0	5 (7)	210 (16)	
21.0–22.9	0 (0)	73 (7)		NA	0 (0)	8 (1)	
≥23.0	0 (0)	30 (3)		Mean	10.3	11.4	
NA	1 (3)	8 (1)					
Mean	13.5	17.3					

*χ^2 test.
†Fisher's exact test.
NA, not accessible; YUMC, Yonsei University Medical Center.

examination, fewer than those reported in European studies, with an incidental diagnoses of 15–28%. As almost all tumours were found after symptom development, delays in the diagnosis were certain to have occurred.

We investigated the survival rate of choroidal melanoma patients who were treated with Ru-106 brachytherapy, and the estimated 5-year overall and disease-specific survival rates were 84% and 87.7%, respectively. This result is in accordance with previous reports on Ru-106 brachytherapy for choroidal melanoma, which estimated the overall 5-year survival rate at 79.6–84.0%.[12–15] In fact, our study is difficult to compare with other studies on brachytherapy because there was a large proportion of patients (14 patients, 23%) with large tumours and various adjunctive treatments. Although there were 21 patients who were treated with brachytherapy combined with tumour excision, whether excision was conducted or not did not affect the survival or metastasis rate. Moreover, previous studies have revealed that endoresection or transscleral tumour resection combined with brachytherapy in large tumours had similar survival outcomes to brachytherapy alone.[16 17] Therefore, although we treated many large choroidal melanomas with combined tumour excision, the survival rate of Korean patients treated with brachytherapy seems to be comparable to that of Caucasians.

Thirteen (21%) out of 61 patients who were primarily treated with brachytherapy eventually underwent enucleation, and the estimated 5-year enucleation rate was 25.4%. This enucleation rate is slightly higher than the 12.5–18.0% reported previously.[13 18] This difference may have arisen because the mean height of choroidal melanomas in our study was higher than in these reports. Tumour height was a significant risk factor that increased the enucleation rate even in multivariate analyses in this study (HR 1.3, p=0.019). It is possible that the higher enucleation rate of this study is due to additional tumour resection treatments, which have complications that can lead to phthisis. However, previous reports have found no difference in eye retention rates between brachytherapy and trans-scleral tumour resection,[16] and the enucleation rate after endoresection was

Table 3 Variables measured as being significantly related to outcomes using univariate Cox proportional hazard analysis

Outcome event	Significant variables	p value	HR (95% CI)
Overall (111 patients)			
Metastasis	LBD	<0.001	1.4 (1.2 to 1.6)
	Height	<0.001	1.3 (1.1 to 1.5)
	Ciliary body involvement	0.02	3.7 (1.6 to 8.5)
Metastatic death	LBD	0.01	1.3 (1.1 to 1.6)
	Height	0.002	1.4 (1.1 to 1.6)
	Ciliary body involvement	0.001	5.3 (1.9 to 14.9)
Brachytherapy (61 patients)			
Metastasis	LBD	0.010	1.4 (1.1 to 1.7)
	Height	0.048	1.3 (1.0 to 1.6)
	Secondary enucleation	0.039	3.7 (1.1 to 12.8)
Metastatic death	LBD	0.032	1.4 (1.0 to 2.0)
	Height	0.050	1.4 (1.0 to 1.9)
Enucleation	Height	0.012	1.3 (1.1 to 1.6)
	Local recurrence	<0.01	12.5 (4.2 to 37.6)

LBD, largest basal diameter

Table 4 Variables measured as being significantly related to outcome events using multivariate Cox proportional hazard analysis

Outcome event	Significant variables	p value	HR (95% CI)
Overall (111 patients)			
Metastasis	Height	0.012	1.2 (1.1 to 1.5)
Metastatic death	Height	0.049	1.2 (1.0 to 1.5)
Brachytherapy (61 patients)			
Metastasis	None		
Metastatic death	NA		
Enucleation	Height	0.019	1.3 (1.0 to 1.7)
	Local recurrence	<0.01	12.1 (3.9 to 37.2)

NA, not accessible.

relatively low (5.8% in 70 months).[17] Furthermore, additional tumour resection treatments were not found to be a significant risk factor for increasing the enucleation rate after brachytherapy in this study.

The higher enucleation rate was likely to have resulted from tumours receiving an insufficient dose of radiation. After Ru plaques to treat choroidal melanoma were first introduced in the 1960s by Lommatzsch,[19] Ru plaques have typically been used for tumours with a maximal thickness of 7.0 mm[20] in order to deliver 100 Gy in total to the tumour apex as Lommatzsch first suggested.[12] Although we always administered the Ru brachytherapy as a combined treatment with TTT, there were many patients (14 in all) who had a reduced apex dose due to their thick tumour height, and whether the apex dose achieved 85 Gy or not significantly affected the enucleation rate in 40 patients treated with brachytherapy. (p=0.01, log-rank test). The enucleation rate may have reduced if we had selected Ru plaques only for the patients with tumour heights less than 7 mm.

Despite the larger height, which was found to be a significant risk factor for metastasis or metastatic death in this study, Koreans have similar prognoses when compared to Caucasians. We assume that similar prognoses are because of the benign characteristics of choroidal melanoma in Koreans. First, the tumour regression rate after brachytherapy is lower than in Caucasians. In 40 patients who were treated with brachytherapy only, the median tumour height was 5.9 mm before treatment and tumours regressed to 69.2% of the original thickness (mean height, 4.1 mm) at 18 months after brachytherapy. More rapid and higher percentages of tumour regression after radiotherapy predicted a poor prognosis because mitotically and metabolically active tumours are more responsive to radiotherapy.[21] Previous studies that investigated the dynamics of tumour height after brachytherapy showed that the height of choroidal melanomas stabilised to 61% of the initial height on average after approximately 18–24 months following treatment.[22] Therefore, choroidal melanomas in Korean patients may have less metabolically active tumours with a more favourable prognosis. Secondly, the favourable prognosis could also be attributable to the relatively low rate of chromosome 3 monosomy in Korean patients. Previous research conducted in our institute has revealed that Koreans have lower levels of monosomy 3, which is known to correlate to poor prognosis in choroidal melanoma.[23]

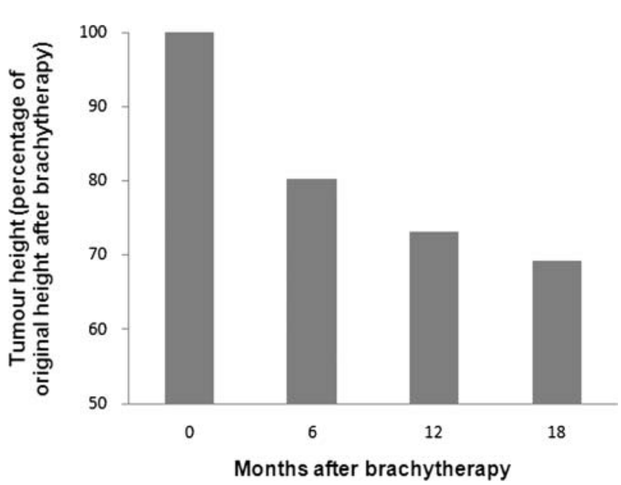

Figure 1 Regression rate in tumour height (as a percentage of initial tumour height after combined Ru-106 brachytherapy and trans-pupillary thermotherapy).

In conclusion, choroidal melanomas in Korean patients have a tendency to grow vertically with relatively large heights and small LBDs. The prognosis of choroidal melanomas overall and their prognosis following Ru brachytherapy appear to be similar to the findings of previous studies in Caucasian populations. The enucleation rate after brachytherapy in Korean patients seems to be higher, but the high enucleation rate may have been related to greater initial tumour height, which could have caused the radiation dose to be insufficient for controlling tumours. Because the follow-up period was relatively short in this study, long-term observation will be required in the future.

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